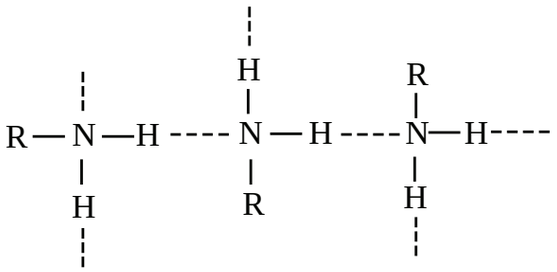

The boiling point of various amines follows the order,

$$CH_3 - \ddot{N}H_2 > (CH_3)_2\ddot{N}H > (CH_3)_3\ddot{N}$$

$$1^{\circ} > 2^{\circ} > 3^{\circ}$$

Amines have lower boiling point than alcohols because nitrogen has lower electronegative value than oxygen and hence the N-H bond is less polar than -OH bond.

Table Boiling points of amines, alcohols and alkanes of comparable molecular weight.

S.NO.	Compound	Molecular mass	Boiling point (K)
1.	$CH_3(CH_2)_2NH_2$	59	321
2.	C_2H_5-NH-CH_3	59	308
3.	$(CH_3)_3N$	59	277
4.	$CH_3CH(OH)CH_3$	60	355
5.	$CH_3CH_2CH_2CH_3$	58	272.5

3) Solubility

Lower aliphatic amines are soluble in water, because they can form hydrogen bonds with water molecules. However, solubility decreases with increase in molecular mass of amines due to increase in size of the hydrophobic alkyl group. Amines are insoluble in water but readily soluble in organic solvents like benzene, ether etc.

13.2.5 Chemical properties

The lone pair of electrons on nitrogen atom in amines makes them basic as well as nucleophilic. They react with acids to form salts and also react with electrophiles.

They form salts with mineral acids

Example:

$$C_6H_5 - \ddot{N}H_2 + HCl \rightleftharpoons C_6H_5 - \overset{+}{N}H_3Cl^-$$

Aniline — Anilinium chloride

Expression for basic strength of amines

In the aqueous solutions, the following equilibrium exists and it lies far to the left, hence amines are weak bases compared to NaOH .

XII U13-Organic Nitrogen Compounds.indd 211 11-01-2021 16:30:30